

Descriptive analysis of harmonized broadband data

This document provides a brief descriptive overview of the harmonized German municipal broadband panel dataset (2005-2021).

Last updated: May 7, 2026. Plots were regenerated from `output/panel_data_public.csv`.

Data quality notes

Use with caution

This dataset has known limitations that users should be aware of:

1. **Methodological breaks:** Two major breaks (2010, 2015) affect comparability across years
2. **Missing years and tiers:** 2009 is absent from the municipality panel; 2005-2008 only report the historical baseline tier
3. **Filtered observations:** Step 06 filters 68,328 post-deduplicated long rows (2.22%) due to unmapped AGS codes (see [unmapped_ags_documentation.md](#) for details)

If using prior data

If you previously used this dataset, please re-download. Fixes include:

- 2005-2008 baseline coverage now correctly populated (~82% mean, was 0%)
- All values validated to [0, 100] range without capping invalid percentages
- AGS codes validated against official Destatis 2021 reference
- Missing speed tiers preserved as missing values rather than filled with zero

Discontinuous changes in the data

The panel data is constructed from historical sources with changing methodologies. This results in two “breaks” in the time series that users must be aware of.

The 2009/2010 break

To create a continuous series for basic internet availability, the `share_broadband_baseline` variable was constructed from two different underlying sources:

- **For 2005-2008:** The variable is based on historical DSL availability data (`speed_mbps_gte = 0.128`, representing ≥ 0.128 Mbps).
- **For 2009:** No municipality panel is available in the current input files.
- **From 2010 onwards:** The variable is based on ≥ 1 Mbps coverage. Since ≥ 1 Mbps implies ≥ 0.128 Mbps, this is a lower bound on basic access.

The consequence is a discontinuity in the data where the definition of the baseline metric changes. In the current panel, this appears as a sharp drop between 2008 and 2010, when the source switches from historical DSL availability to the stricter ≥ 1 Mbps metric. This is distinct from organic change in broadband availability.

The 2015 break

As noted in the project documentation, 2015 marks a major structural break due to a change in the primary data provider and reporting standards. This affects all speed tiers and is visible as a very large, discontinuous jump in coverage levels. The `method_change_2015` dummy variable is included in the dataset to allow researchers to control for this event.

Descriptive plots

Average annual coverage

This plot shows the mean coverage for the baseline metric alongside higher speed tiers. The two key events are clearly visible: the **2010 definition break** in the baseline series and the large **2015 methodological break**.

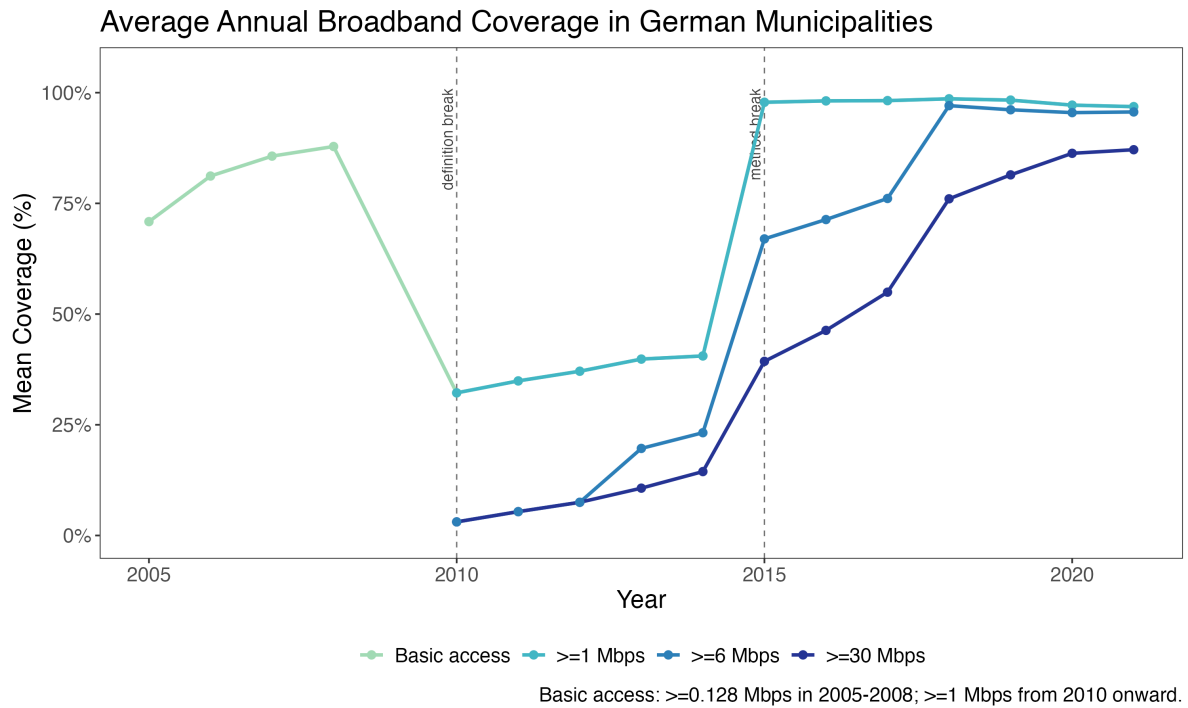


Figure 1: Average Annual Coverage

Year-over-year change in coverage

This plot highlights the magnitude of the two breaks. The **baseline** change between 2008 and 2010 is substantial and definition-driven. The 2015 spike is large across all speed tiers and reflects the documented methodological break.

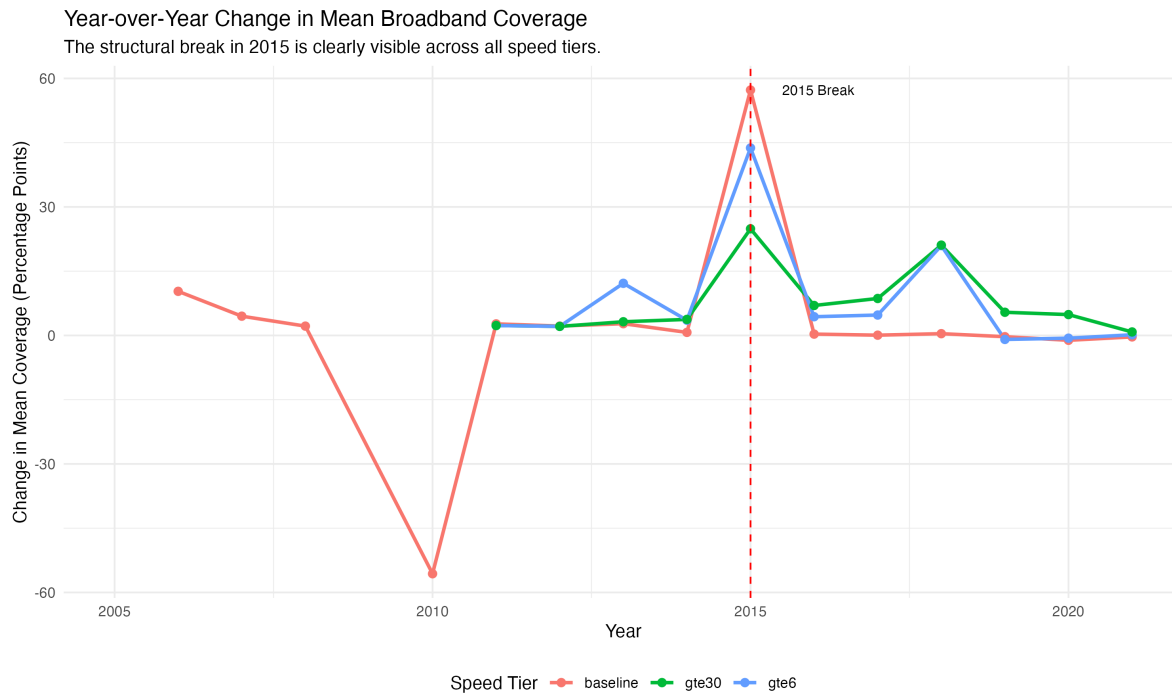


Figure 2: Year-over-Year Change in Coverage

Distribution of coverage levels

This set of histograms shows the distribution of coverage values. The `share_broadband_baseline` is heavily skewed towards 100%, indicating that most municipalities achieved full basic coverage relatively early. In contrast, higher speed tiers show more variation and a larger concentration near 0%.

Distribution of Municipality-Level Broadband Coverage
Shows high concentration at 0% and near 100% coverage.

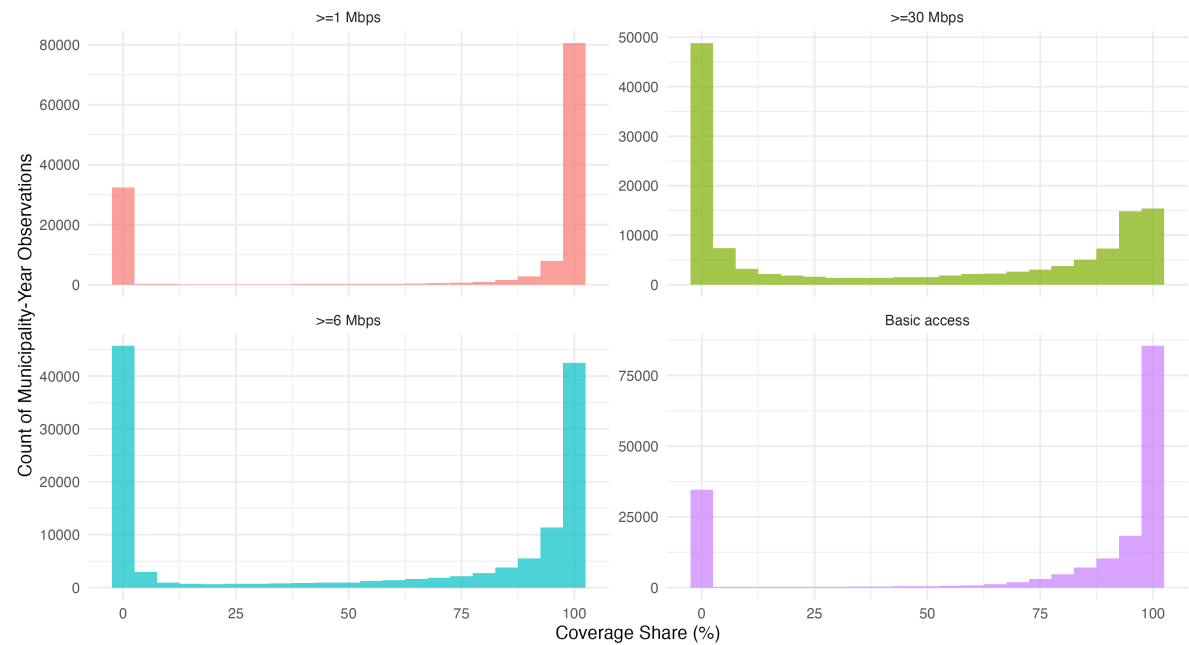


Figure 3: Distribution of Coverage Levels